CODING FOR DATA SCIENCE AND ANALYTICS

Exercise Session 2: Booleans, Conditions and Loops

Exercise 1 — Truth Tables

Create truth tables for the AND, OR, NOT logical operations:

- `A and B`: true when both conditions are true.
- `A or B`: true when at least one condition is true (also when both are true).
- `not A`: reverses the Boolean value.

A	B	A and B
True	True	?
True	False	?
False	True	?
False	False	?

A	B	A or B
True	True	?
True	False	?
False	True	?
False	False	?

A	not A
True	?
False	?

First approach

- Use Boolean operands.
- Repeated blocks are fine here. We will automate the table after practising loops.

Exercise 2 — Grades Conversion

Assume you are given a student's numerical grade and their class attendance percentage. Create a program that determines their letter grade and pass/fail status based on the following rules:

Letter Grade:

- A: 90-100
- B: 80-89
- C: 70-79
- D: 60-69
- F: Below 60

Pass/Fail determination:

- Pass: Grade ≥ 60 **and** Attendance $\geq 75\%$
- Fail: Grade < 60 **or** Attendance $< 75\%$

Exercise 3 — Spelling a Word

Take a word as input from the user and print each letter on a new line.

Example input

```
Python
```

Expected output

```
P
y
t
h
o
n
```

Hint

Use the `input()` function to get the word, and a `for` loop to iterate through the characters of the string.

Exercise 4 — Fibonacci Sequence

Calculate and print the first 20 numbers in the Fibonacci sequence.

The sequence is defined by the following formula:

$$F_0 = 0, \qquad F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2$$

Expected output start:

```
0, 1, 1, 2, 3, 5, 8, ...
```

Exercise 5 — Even or Odd

A - Ask the user for a number and return whether it is even or odd.

Hint

Use the `input()` function and the modulo operator (`%`).
Assume the user enters a valid value.

B - Modify the program so that it uses a `while` loop to continue asking the user for numbers until they type “exit”.

Exercise 6 — Questionnaire Data Quality

A string contains 10 answers to a multiple-choice question: `A`, `B`, and `C` are valid answers; `?` represents a missing answer.

```
answers = "ABAC?BBCA?"
```

1. Count the missing answers.
2. Count the occurrences of `A`, `B`, and `C` separately.
3. Calculate the percentage of missing answers.
4. Print `"Data to review"` if this percentage exceeds 15%; otherwise, print `"Acceptable quality"`.

Hint

Use a `for` loop, `if/elif/else`, and counters initialised before the loop. Assume the string always contains exactly 10 characters, each being `A`, `B`, `C`, or `?`.